
Remark. Different non-decreasing functions can generate the same measure. Consider $f = \mathbb{1}_{[0, \infty)}$ and $g = \mathbb{1}_{(0, \infty)}$. Then $f(0) \neq g(0)$. Take $a < b$ and consider the following cases

$a, b \leq 0$) Both $f(a) = 0$ and $g(b) = 0$, thus $\mu_f((a, b]) = \mu_g((a, b]) = 0$.

$a, b \geq 0$) Since $0 \leq a < b$, then $f(a) = f(b) = 1$ and $g(a) = g(b) = 1$, thus $\mu_f((a, b]) = \mu_g((a, b]) = 0$.

$a < 0 < b$) Note that

$$\mu_f((a, b]) = f(b) - f(a) = 1 - 0 = 1$$

$$\mu_g((a, b]) = g(b) - g(a) = 1 - 0 = 1$$

Therefore $\mu_f = \mu_g$ despite $f \neq g$.

Theorem 0.1.

If μ is a Borel measure that is finite for all bounded intervals, then there is a unique (up to an additive constant) non-decreasing right-continuous function $F : \mathbb{R} \rightarrow \mathbb{R}$ such that

$$\mu((a, b]) = F(b) - F(a), \quad a < b.$$

Proof. Define the candidate generating function

$$F(x) := \begin{cases} \mu((0, x]) & x \geq 0 \\ -\mu((x, 0]) & x < 0 \end{cases}.$$

Since $\mu((x, y]) < \infty$ for any $x, y \in \mathbb{R}$, it follows that $F(x) < \infty$ for all x . Suppose then $a < b$ and consider the following cases

$a, b \geq 0$) Since $(a, b] = (0, b] \setminus (0, a]$, then

$$F(b) - F(a) = \mu((0, b]) - \mu((0, a]) = \mu((a, b]).$$

$a, b < 0$) Since $(a, b] = (a, 0] \setminus (b, 0]$, then

$$\begin{aligned} F(b) - F(a) &= (-\mu((b, 0])) - (-\mu((a, 0])) \\ &= \mu((a, 0]) - \mu((b, 0]) \\ &= \mu((a, b]) \end{aligned}$$

$a < 0 \leq b$) Since $(a, b] = (a, 0] \dot{\cup} (0, b]$, then

$$\begin{aligned} F(b) - F(a) &= \mu((0, b]) - (-\mu((a, 0])) \\ &= \mu((0, b]) + \mu((a, 0]) \\ &= \mu((a, b]) \end{aligned}$$

Clearly F is non-decreasing since $F(y) - F(x) = \mu((x, y]) \geq 0$ for $y > x$. Take $x \in \mathbb{R}$ and $(h_n)_{n \in \mathbb{N}}$ with $h_n \geq 0$ and $\lim_{n \rightarrow \infty} h_n = 0$. Note then that

$$F(x + h_n) - F(x) = \mu((x, x + h_n])$$

which by continuity from above means $\lim_{n \rightarrow \infty} \mu((x, x + h_n]) = 0$ and hence $\lim_{n \rightarrow \infty} F(x + h_n) = F(x)$. Thus F is right-continuous.

For uniqueness, suppose there is some G that is also non-decreasing and right-continuous such that for $a < b$

$$\mu((a, b]) = G(b) - G(a).$$

Then for all $a < b$

$$F(b) - F(a) = G(b) - G(a) \implies F(b) - G(b) = F(a) - G(a).$$

This means that $F - G$ is a constant function, hence $G = F + c$ for some constant c . ◇